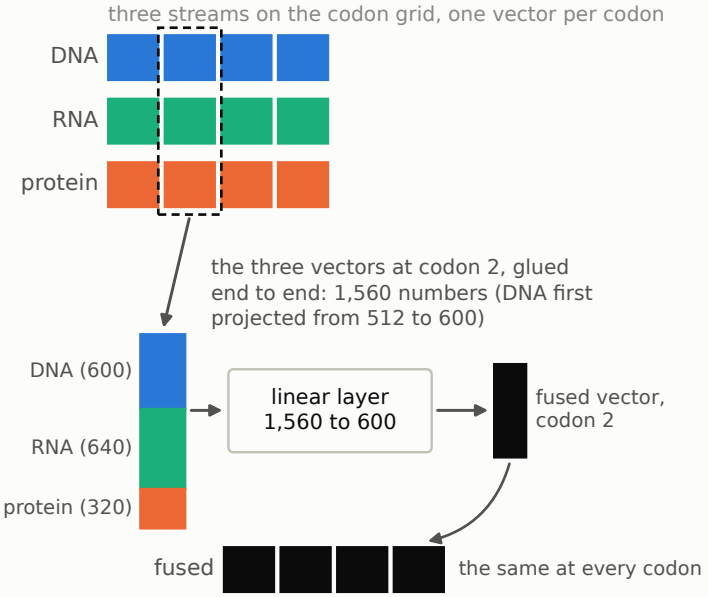
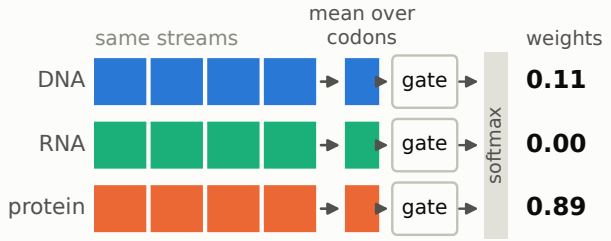
a Concatenation: keep everything

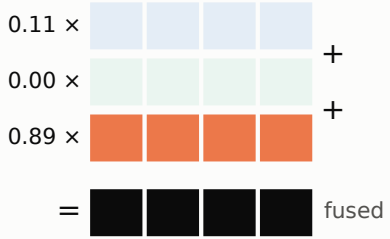


Nothing is discarded. The TextCNN that follows decides which features of which modality to use.

b Gated attention: three weights

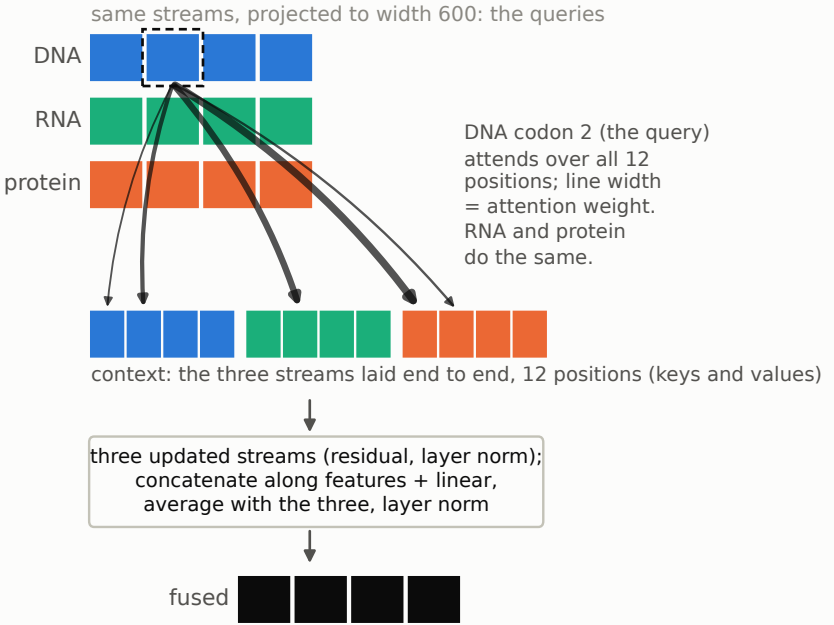


the three weights multiply the full streams, codon by codon:



Three numbers per sequence, decided from averages and applied at every codon; the entropy term pushes them toward one-hot. With RNA at 0.00 the whole RNA stream is dropped.

c Cross-modal attention: read anything anywhere



Every position can combine features from any modality at any codon: the most expressive head, and the only one that never lost to the best single encoder in Phase A.